

Michael Seo

CONTACT INFORMATION	Personal Website: mikejseo.github.io	Email: mikejseo7[at]gmail.com
RESEARCH INTERESTS	causal inference, indirect treatment comparison, network meta-analysis, Bayesian methods	
EDUCATION	Ph.D Biostatistics and Epidemiology, University of Bern, 2019-2022 M.S Statistics, Stanford University (GPA: 3.7), 2012-2014 B.S Statistics, Graduation with High Distinction, Duke University (GPA: 3.7), 2007-2011	
EMPLOYMENT	Principal Data Scientist, Real World Evidence (RWE) , Johnson & Johnson Innovative Medicine, 2026-Present Access Evidence Lead (HTA Statistician) , Roche, 2022-2026 Ph.D Student in Biostatistics and Epidemiology , Institute of Social and Preventive Medicine, University of Bern, 2019-2022 Biostatistician , LLX Solutions, 2018-2019 Research Associate , Department of Biostatistics, Brown University, 2015-2017 Quantitative Analyst , In4mation Insights, 2015 Summer Research Assistant , Department of Statistics, Stanford University, 2014-2015 Research Assistant , Children's Environmental Health Initiative, Duke University, 2011-2012	
PUBLICATIONS	Efthimiou, O., Chalkou, K., Seo, M. , Pellegrini, F., Salanti, G., and Debray, T. (2026). Modeling personalized treatment effects using multiple data sources. <i>Comparative Effectiveness and Personalized Medicine Research Using Real-World Data</i> . Chapter 18, 464-484. CRC Press. Efthimiou O, Seo M , Chalkou K, Debray T, Egger M, Salanti G (2024). Developing clinical prediction models: a step-by-step guide. <i>BMJ</i> , 386. Seo M , Furukawa TA, Karyotaki E, Efthimiou O (2023). Developing prediction models when there are systematically missing predictors in individual patient data meta-analysis. <i>Research Synthesis Methods</i> , 14, 455-467. Efthimiou O, Hoogland J, Debray TPA, Seo M , Furukawa TA, Egger M, White IR (2023). Measuring the performance of prediction models to personalize treatment choice. <i>Statistics in Medicine</i> , 42, 1188-1206. Yoshida K, Seo M , Luo Y, Sahker E, Cipriani A, Leucht S, Iwatsubo T, Efthimiou O, Furukawa TA (2022). Personalized prediction of Alzheimer's disease and its treatment effects by donepezil: an individual participant data meta-analysis of eight randomized controlled trials. <i>Journal of Alzheimer's disease</i> , 89, 1143-1157. Seo M , Debray TPA, Ruffieux Y, Gsteiger S, Bujkiewicz S, Finckh A, Egger M, Efthimiou O (2022). Combining individual patient data from randomized and non-randomized studies to predict real-world effectiveness of interventions. <i>Statistical Methods in Medical Research</i> , 31, 1355-1373. Efthimiou O, Seo M , Karyotaki E, Cuijpers P, Furukawa TA, Schwarzer G, Rücker G, Mavridis D (2022). Bayesian models for aggregate and individual patient data component network meta-analysis. <i>Statistics in Medicine</i> , 41, 2586-2601. Seo M , White IR, Furukawa TA, Imai H, Valgimigli M, Egger M, Zwahlen M, Efthimiou O (2021). Comparing methods for estimating patient-specific treatment effects in individual patient data meta-analysis. <i>Statistics in Medicine</i> , 40, 1553-1573. Seo M , Furukawa TA, Veroniki AA, Pillinger T, Tomlinson A, Salanti G, Cipriani A, Efthimiou O (2021). The Kilim plot: A tool for visualizing network meta-analysis results for multiple outcomes. <i>Research Synthesis Methods</i> , 12, 86-95.	

Furukawa TA, Debray T, Akechi T, Yamada M, Kato T, **Seo M**, Efthimiou O (2020). Can personalized treatment prediction improve the outcomes, compared with the group average approach, in a randomized trial? Developing and validating a multivariable prediction model in a pragmatic megatrial of acute treatment for major depression. *Journal of Affective Disorders*, 274, 690-697.

Khan MS, Khan AR, Khan AI, **Seo M**, Yasmin F, Usman MS, Moustafa A, Schmid CH, Kalra A, Ikram S (2020). Comparison of revascularization strategies in patients with acute coronary syndrome and multivessel coronary disease: A systematic review and network meta-analysis. *Catheterization and cardiovascular interventions*, 96, E447-E454.

R PACKAGES Gregory Chen, **Michael Seo** and Isaac Gravestock (2024). maicplus: Matching Adjusted Indirect Comparison. R package version 0.1.0. <https://hta-pharma.github.io/maicplus/>.

Michael Seo (2022). bipd: Bayesian Individual Patient Data Meta-Analysis using 'JAGS'. R package version 0.3. <https://CRAN.R-project.org/package=bipd>.

Michael Seo and Christopher Schmid (2020). bnma: Bayesian Network Meta-Analysis using 'JAGS'. R package version 1.5.0. <https://CRAN.R-project.org/package=bnma>.

Robert Tibshirani, **Michael Seo**, Gil Chu, Balasubramanian Narasimhan and Jun Li (2018). samr: Significance Analysis of Microarrays. R package version 3.0. <https://CRAN.R-project.org/package=samr>.

CONTRIBUTED PRESENTATIONS The subtle yet impactful choices in matching-adjusted indirect comparison for time-to-event endpoints: insights from simulation (poster). Annual Conference of the International Society for Clinical Biostatistics (ISCB), Basel; August 2025.

Developing prediction models when there are systematically missing predictors in an individual patient data meta-analysis. Joint Statistical Meeting (JSM), Seattle; August, 2021.

bnma: Bayesian Network Meta-Analysis using 'JAGS'. Evidence Synthesis and Meta-Analysis in R Conference; January, 2021.

Predicting real world effectiveness of interventions, combining individual patient data from multiple randomized and non-randomized studies. Annual Conference of the International Society for Clinical Biostatistics (ISCB), Krakow; August, 2020.

The Kilim plot: a tool for visualizing network meta-analysis results for multiple outcomes (poster). Annual Conference of the International Society for Clinical Biostatistics (ISCB), Krakow; August, 2020.

Comparing methods for variable selection in individual patient data meta-analysis. XXXIst Conference of the Austro-Swiss Region (ROeS) of the International Biometric Society, Lausanne; September, 2019.

TEACHING ASSISTANTSHIP

Applied Logistic Regression, University of Bern, 2021

Prognostic Research: from Basics to Modelling, University of Bern, 2020, 2021

Evidence Synthesis Methods, University of Bern, 2019

Applied Regression Analysis (PHP 2511), Brown, Spring 2016

Fundamentals of Probability and Statistical Inference (PHP 2515), Brown, Fall 2015

Probability and Statistics Inference (STAT 103), Duke, Spring 2011

Statistics (STAT 114), Duke, Fall 2010

Probability and Statistics for Engineers (STAT 113), Duke, Spring 2010

Probability (STAT 104), Duke, Fall 2009

PROGRAMMING R, Python, SAS, SQL

LANGUAGES Korean, English